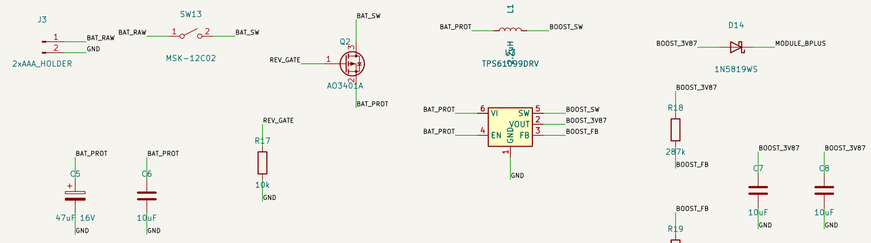
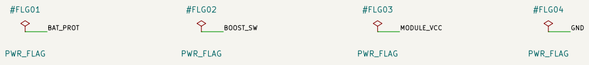
1 POWER — REPLACEABLE 2xAAA, NO AAA CHARGING

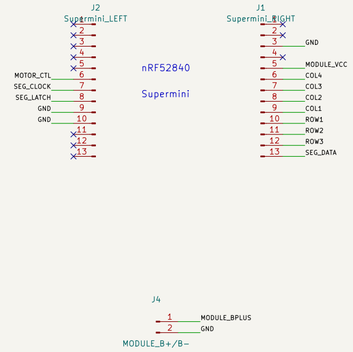


U3 output = 1.0 V × (1 + 287k/100k) = 3.87 V

D14 blocks onboard USB charger from feeding the AAA cells.



2 NRF52840 SUPERMINI

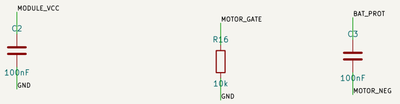


J4 represents B+ / B− pads on the module; verify board variant.

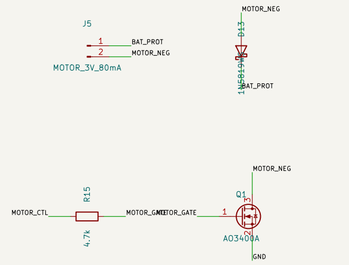
J1.5 VCC is an OUTPUT from the module.

3 TWO CASCADED 74HC595





5 VIBRATION MOTOR

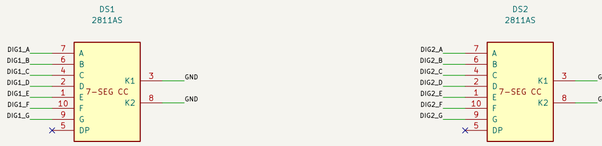


D13 cathode to BAT_PROT: motor gets raw protected AAA voltage.

4 4x3 KEY MATRIX — 7 GPIO, no 74HC165

D1–D12 remove phantom keys when several buttons are pressed.

30 ms button debounce is implemented in firmware.



2811AS pinout: A7 B6 C4 D2 E1 F10 G9 DP5 K3/K8

Firmware must use active-HIGH segment bits for common-cathode LEDs.

Schematic review: module and mechanical footprints require physical check
12-key 4x3 matrix; two common-cathode 2811AS displays

Sheet: /
File: StudentTag.kicad_sch

Title: StudentTag — 2xAAA BLE clicker

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/1